

Model IR-LEL

Combustible Gas Sensor



Scrolling Full Message/Text Display



Non-dispersive Infrared Optical
(shown as PN KSA-9H7-214520-100C in SS junction box)



Description

Model IR-LEL is a non-intrusive "Smart" sensor designed to detect and monitor combustible hydrocarbon gases in air over the range of 0-100% LEL using miniature non-dispersive Infrared Optical (NDIR) sensor technology. As compared to catalytic bead sensors, with NDIR there is no risk of sensor poisoning, no risk of high concentration saturation, and no need for O₂ to be present. The plug-in, field replaceable detector features oversized gold-plated connections that help prevent corrosion. The IR-LEL's rugged framework includes an electro-polished 316 stainless steel housing with fully encapsulated electronics and dual layer surge protection. This innovative design virtually eliminates sensor failure due to water ingress, corrosion, vibration, and transient spikes. A primary feature of the Model IR-LEL is embedded intuitive software that simplifies operator interface by guiding the user through routine calibration, configuration, and fault diagnostic functions using a built-in alpha/numeric display. The Model IR-LEL is equipped with standard analog 4-20mA, and HART outputs.

Teledyne Gas and Flame Detection's combustible gas sensor has an infinite shelf life and is supported by a 5-year warranty.

Applications

- Oil and Gas Drilling Rigs
- Oil and Gas Production Sites
- Offshore Production Platforms
- FPSOs
- LNG/CNG Plants
- Refining and Petrochemical
- Pulp and Paper Mills
- Waste Water Treatment Plants

Gas Detected

Methane	Hexane
Ethane	Ethylene
Propane	Gasoline Vapor
Butane	Diesel Fuel Vapor
Pentane	Jet Fuel Vapor

For other gases consult the factory.

Features

Failsafe User-Friendly Interface

- LED Display (With Antiglare Cover)
- Full Text Display Method
- Non-intrusive Interface
- Auto Zero/Auto Span
- Pre-emptive Fault Diagnostics

Environmentally Bulletproof

- Electropolished 316SS Construction
- 100% Epoxy Encapsulated Circuitry
- Bulletproof I/O Protection
- Water-Proof, Corrosion-Proof, Vibration-Proof

Modular and Serviceable

- Modular Design
- Plug and Play Components
- Quick Thread Release (For Sensor Replacement)
- Integral Calibration Port

Model IR-LEL

Combustible Gas Sensor

System Specifications

Sensor Type

Continuous diffusion/adsorption
Non-dispersive Infrared Optical (NDIR) - Combustible Gas
Sub-miniature plug-in field replaceable

Sensor Life

5 years typical

Measurement Range

0-100% LEL (lower explosive limit)
0-100% by volume

Accuracy

±2% F.S. (0-50% LEL)

Repeatability

±2% F.S.

Response Time*

T50 < 6 seconds, T90 < 10 seconds

Zero Drift (typical)

≤2% per 2 years

Outputs

Linear 4-20 mA DC
HART

Hazardous Location Approvals

Explosion proof
cCSAus (80209826)
Class I, Division 1, Groups B, C, D (Tamb = -40°C to +75°C)
Class I, Zone 1, Group IIB+H2
IECEX (DEK 24.0040X)
Ex db IIB+H2 T4 Gb (Tamb = -40°C to +70°C)

Ingress Protection

NEMA 4X, IP66

Safety Approvals

cCSAus Performance to ANSI/ISA 60079-29-1 (12.13.01)-2013 and CSA
C22.2 No. 152-M1984
IECEX

Warranty

Plug-in detector - 5 years
Transmitter - 2 Years

* As tested with 1 lpm flowrate

Environmental Specifications

Operating Temperature Range

-40°F to +167°F; -40°C to +75°C

Storage Temperature Range

-40°F to +167°F; -40°C to +75°C

Operating Humidity Range

0-100% RH non-condensing

Specifications subject to change without notice

Mechanical Specifications

Dimensions

11"H x 6.1"W x 3.75"D; 280mmH x 155mmW x 96mmD
(with junction box)
Mounting holes (J-box) 5.5"; 140mm center to center

Weight

9 lbs; 4.08kg (w/stainless steel j-box)

Electrical Specifications

Power Input

11-30 VDC

Power Consumption

Normal operation = 68mA (<1.7 watt)
Maximum = 85mA (2 watts)

Inrush Current

0.67A @ 24V

RFI/EMI Protection

Complies with EN50270

Analog Output

Linear 4-20mA DC (1,000 ohms max loop load @ 24VDC)

0mA	All Fault Diagnostics
2mA	In-Calibration
4-20mA	0-100% full-scale
22mA	Over-range condition

Status Indicators

4-digit LED display with gas concentration
Full-script menu prompts for AutoSpan,
Set-up Options, and Fault Reporting

Faults Monitored

Loop, Input Voltage,
Zero, Sensor,
Processor, Memory, Calibration

Cable Requirements

Power/Analog
3-wire shielded cable
Maximum distance is 13,300 feet with 14 AWG

I/O Protection

Over-voltage, Miswiring, EMI/RFI Immunity

Integration Options

HART Integration Module (HART Communication Protocol version 7.0, HART Registered)

Order Guide

Specify junction box options (below)

PN KSA-9H7-214520-100C IR-LEL Sensor Assembly with HART and 316 SS J-Box



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IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION IEC Certification System for Explosive Atmospheres

for rules and details of the IECEx Scheme visit www.iecex.com

Certificate No.:	IECEx DEK 24.0040X	Page 1 of 4	<u>Certificate history:</u>
Status:	Current	Issue No: 1	Issue 0 (2025-01-29)
Date of Issue:	2025-07-11		
Applicant:	Teledyne Gas & Flame Detection 14880 Skinner Road Cypress TX 77429 United States of America		
Equipment:	Gas detection sensor assemblies		
Optional accessory:			
Type of Protection:	Ex db		
Marking:	Ex db IIB + H ₂ T4 Gb		

Approved for issue on behalf of the IECEx
Certification Body:

R. Schuller

Position:

Certification Manager

Signature:
(for printed version)

Date:
(for printed version)

2025-07-11

1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
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6825 MJ Arnhem
Netherlands





IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0040X**

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Date of issue: 2025-07-11

Issue No: 1

Manufacturer: **Teledyne Gas & Flame Detection**
14880 Skinner Road
Cypress TX 77429
United States of America

Manufacturing
locations: **Teledyne Detcon, Inc.**
14880 Skinner Road
Cypress, TX 77429
United States of America

**Industrial Detection Solutions
Company (IDS)**
F11 & F12, Dammam 913 Street,
Building No. 2857
MODON-2
Dammam Industrial City 2 34325
Saudi Arabia

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended

STANDARDS :

The equipment and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards

[IEC 60079-0:2017](#) Explosive atmospheres - Part 0: Equipment - General requirements
Edition:7.0

[IEC 60079-1:2014](#) Explosive atmospheres - Part 1: Equipment protection by flameproof enclosures "d"
Edition:7.0

This Certificate **does not** indicate compliance with safety and performance requirements other than those expressly included in the Standards listed above.

TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in:

Test Report:

[NL/DEK/ExTR24.0045/01](#)

Quality Assessment Reports:

[NL/DEK/QAR24.0012/00](#)

[US/ETL/QAR20.0010/05](#)



IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0040X**

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Date of issue: 2025-07-11

Issue No: 1

EQUIPMENT:

Equipment and systems covered by this Certificate are as follows:

FP-LEL Combustible Gas Sensor, IR-LEL Combustible Gas Sensor or IR-CO2 Carbon Dioxide Sensor, and TP-H2S Hydrogen Sulfide Sensor, for fixed installation, designed to detect and monitor combustible gases, CO₂ or H₂S in air.

Ambient temperature range -40 °C to +70 °C

Electrical data

Input rating: 11 to 30 Vdc, 85 mA max.

Output: 4 to 20 mA, RS-485 ModBus.

SPECIFIC CONDITIONS OF USE: YES as shown below:

1. The sensor shall be installed on a metallic grounded structure such as the final assembly assures that sensor enclosure is bonded to ground.
2. Proper precautions shall be taken during installation, operation, maintenance and service of the product in order to avoid the build-up of static charges on the plastic splashguard.
3. The flameproof joints are not intended to be repaired.
4. The screws holding down the retaining plate are special fasteners of type: stainless steel, Philips Pan-head Machine screws, M3 x 0.5 6g/6H having yield strength greater than 40,000 PSI.



IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0040X**

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DETAILS OF CERTIFICATE CHANGES (for issues 1 and above)

Update of the company name for the additional manufacturing location.